

ISRA AMAN AZIZ

isra.aman20@gmail.com | [linkedin.com/in/israamanaziz](https://www.linkedin.com/in/israamanaziz) | github.com/Israamanaziz | israamanaziz.com

PROFESSIONAL SUMMARY

A dynamic bioinformatics graduate with a strong foundation in transcriptomics (RNA-Seq, miRNA-Seq) and metagenomics (16S rRNA analysis using QIIME2), and standard NGS pipelines. Skilled in Python, R, and Bash for data analysis, with exposure to statistical testing, data visualization, and biological interpretation. Experienced in working with public datasets (GEO/SRA) and translating results into structured scientific reports. Actively seeking an opportunity to contribute to genomics research by applying expertise in NGS data analysis, bioinformatics pipelines, and biological data interpretation.

SKILLS

Programming: Python, R, Bash

Bioinformatics & NGS Analysis: RNA-Seq, miRNA-Seq, 16S rRNA (QIIME2, DADA2), DESeq2, FastQC, fastp, STAR, Salmon, SAMtools, PCoA, alpha/beta diversity, miRNA target prediction (miRDB), functional enrichment (KEGG/GO), network analysis (Cytoscape)

Databases: NCBI (GEO, SRA, GenBank), PDB, PubChem, IMPPAT

Tools & Platforms: Galaxy, Git, GitHub, Jupyter Notebook, RStudio, Google Colab, conda/mamba

Wet Lab: PCR, qPCR, agarose gel electrophoresis, microbial culture, spectrophotometry

Research: Experimental design, data analysis, scientific writing, MS Word, MS Excel

RESEARCH INTERESTS

- Diet-gut microbiome interactions in neurobiology
- Neuroinflammation, cognitive function, and brain aging
- Multi-omics and AI/ML for biomarker discovery
- Molecular signatures for neurological and neurodevelopmental disorders

RESEARCH EXPERIENCE

Babasaheb Bhimrao Ambedkar University, Lucknow, India

December 2025 – May 2026

Dissertation Trainee

- Built an end-to-end pipeline for miRNA-Seq analysis using 30+ samples from NCBI GEO datasets for schizophrenia patients
- Performed QC, normalization, and differential expression analysis using DESeq2 in R
- Developed publication-ready volcano, heatmap, and MA plots for gene/miRNA profiling using ggplot2
- Predicted miRNA-mRNA interactions using miRDB
- Implemented supervised ML models, including Random Forest, Logistic Regression, and LASSO, for miRNA-based classification
- Evaluated model performance using ROC-AUC metrics
- Constructed and visualized complex regulatory networks to enhance biological data analysis using Cytoscape

BioinfoCore Solutions Pvt Ltd, Lucknow, India

February 2025 – May 2025

Research Intern

- Performed end-to-end 16S rRNA microbiome analysis using 200+ samples from 6 public GEO datasets
- Investigated gut dysbiosis in Alzheimer's and COVID-19 cohorts using QIIME2 and Galaxy, contributing to insights into disease-associated microbial shifts
- Executed comprehensive data preprocessing, DADA2-based filtering, and taxonomy assignment for large datasets
- Generated publication-ready diversity visualizations, including PCoA, heatmaps, volcano plots, and bar plots, to effectively communicate complex findings
- Prepared a research manuscript based on findings for submission to a peer-reviewed journal

Amity Institute of Biotechnology, AUUP, Lucknow, India

January 2024 – May 2024

Research Intern

- Conducted extensive molecular docking studies and virtual screening of 50 phytocompounds, identifying potential novel ligands targeting breast cancer
- Utilized a suite of bioinformatics tools (PDB, PubChem, IMPPAT, SwissDock, SwissADME, CBDock) to analyze and prioritize bioactive compounds
- Contributed to the computational pipeline for early-stage drug discovery, identifying promising therapeutic candidates

CSIR-National Botanical Research Institute, Lucknow, India

May 2023 – June 2023

Project Trainee

- Investigated and developed a bioinoculant producer (*Azospirillum*) for sustainable agricultural applications
- Performed a range of laboratory techniques, including sterile sampling, culture media preparation, microbial isolation and screening, PCR, AGE, MIC determination, spectrophotometry, and standard curve generation
- Optimized experimental protocols for microbial growth and characterization, contributing to research efficiency

EDUCATION

Babasaheb Bhimrao Ambedkar University, Lucknow, Lucknow, India

2024 – 2026

M.Sc. Bioinformatics

- CGPA: 7.98/10

Amity University Uttar Pradesh, Lucknow, Lucknow, India

2021 – 2024

B.Sc. Microbiology (Honours)

- CGPA: 9.74/10
- Recipient of merit scholarship for 3 consecutive years
- B.Sc. Class Topper

PROJECTS

Plasma Exosomal miRNA Biomarker Discovery Pipeline

- End-to-end bioinformatics and ML workflow to identify candidate plasma exosomal miRNA biomarkers in first-episode schizophrenia
- Built the pipeline from preprocessing, miRNA quantification (miRDeep2), and DESeq2 differential expression through Random Forest/LASSO-based biomarker prioritization, target prediction (miRDB), PPI network analysis (Cytoscape), and GO/KEGG enrichment (clusterProfiler)

Diabetes Prediction using Machine Learning

- Built a Support Vector Machine (SVM) classification model on the Pima Indians Diabetes dataset to predict diabetes risk from clinical features
- Implemented the full ML workflow in Python/scikit-learn: data preprocessing, standardization, train-test splitting, model training, and a basic predictive system

PUBLICATIONS

Under review / In revision:

- [Isra Aman Aziz](#), Prekshi Garg (2026). Computational Microbiome Analysis Identifies Shared Microbial Dysbiosis Indicating Increased Risk of Alzheimer's Disease Post-COVID-19 Infection. [Accepted in a peer-reviewed journal]

Published:

- [Isra Aman Aziz](#), Jyoti Prakash, Rachna Chaturvedi, Ruchi Yadav. Computational Exploration of Natural Compounds as Potential Ligands Targeting Breast Cancer. *Asian Journal of Pharmaceutical Research and Health Care* 16(4): 410–418, Oct–Dec 2024. DOI: 10.4103/ajprhc.ajprhc_173_24
- Pallabi Chatterjee, [Isra Aman Aziz](#), Amarjit Singh, Aditi Singh. Microbiome in Teenagers – Acquisition and Development. In Bentham Science Publishers eBooks, 1–13, 2024. <https://doi.org/10.2174/9789815274431124010003>
- Shreetama Bhattacharjee, [Isra Aman Aziz](#), Aditi Singh. Frangipani: A Common Plant With Antimicrobial Potential Against Common Bacterial Pathogens. *International Journal of Pharmacy and Pharmaceutical Research* 30(4): 313–321, 2024. <https://doi.org/10.25166/ijpr.2024.30.4.21>
- [Isra Aman Aziz](#), Amaan Arif, Saba Hasan. Types of Carcinogens: Physical and Chemical Carcinogens. In *Introduction to Cancer Biology and Genetics: Cancer Genetics Unlocked* (2024). Notion Press.

CONFERENCE PRESENTATIONS & AWARDS

- Poster Presentation, **Neuroscience Research Symposium 2025**, Centre for Brain and Mind (CBM), National Centre for Biological Sciences (NCBS), Bangalore — “Microbiome Analysis Reveals Shared Dysbiosis Linked to Alzheimer's Disease Risk Post-COVID-19”
- Oral Presentation, 3rd International Conference on Recent Advances in Biotechnology and Nanobiotechnology (Int-BIONANO 2024), Amity University Uttar Pradesh — “Furanocoumarins as Potential Anti-Breast Cancer Agents via Structure-Based Docking”
- Oral Presentation, International Conference on Recent Trends in Inter-Disciplinary Innovations Bridging Computer Science and Life Science (RTIICSM 2024), Rathinam College of Arts and Science, Coimbatore — “Computational Screening of Natural Compounds Targeting Breast Cancer”
- Best Poster Award, National Symposium by Translational Biomedical Research Society (TBRS 2023) — “Antioxidant Potential of Selected Indian Plants and Spices”
- Best Poster Award, International Conference on Sustainable Development and Disaster Resilience through Innovation (ICSDDRTI 2023), Amity University Uttar Pradesh — “Antimicrobial Potential of Frangipani Against Bacterial Pathogens”

EXTRACURRICULARS

- Student Coordinator, Entrepreneurship Cell, AUUP — Promoted E-Cell incubation and anchored events at the Incubation Cell (June 2023 – March 2024)
- Student Group Leader, Navreeta Foundation — Advocated for women's rights and menstrual hygiene (Aug 2022 – Jan 2023)
- Volunteer, Hamari Pahchan NGO — Supported education for underprivileged children (Mar 2023 – May 2023)